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RESEARCH ARTICLE

Event-Based Slip Estimation Framework for Space Rovers Traversing Soft Terrains

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ABSTRACT Estimating terrain-induced longitudinal slip poses a significant challenge in space rover navigation, particularly when traversing soft terrains in low-light conditions. Precise estimation of this slip is important for rover navigation algorithms, as it helps to prevent rovers from traversing areas with excessive slippage, thereby avoiding entrapment—an outcome that could lead to mission failure. This paper presents a novel event-based longitudinal slip estimation framework designed to operate in low-light conditions. Existing vision-based slip estimation techniques suffer from low temporal resolution sensors and a high sensitivity to motion blur. These limitations prevent the immediate detection of slip events, which can occur in microseconds, making such methods unsuitable for highly variable terrain environments such as lunar surfaces. Our proposed framework addresses these challenges, offering a robust solution for precise slip estimation in microseconds with limited energy consumption. Specifically, we generate the data directly from real-time wheel traces at varied light conditions and motion patterns and then propose a framework to estimate the slip. This framework consists of a denoising filter and a Hough transformer. Experiments conducted on a single-wheel test rig demonstrate the effectiveness of the proposed event-based framework in estimating slip ratios that range from 2% to 90% with an estimated accuracy of above 95% in well-lit conditions and above 94% in low-light conditions. A supplementary video is available at <https://youtu.be/eac6Q2a8A9g>.

INDEX TERMS Neuromorphic vision sensor, event-based, slip estimation, space rovers, soft terrains, low-light.

I. INTRODUCTION

Safe navigation plays a critical role in the operation of planetary rovers, particularly in the face of challenging and unpredictable terrains encountered in space environments [1]. Such terrains often have features like loose soil, steep inclinations, and craters; thus, identifying potential hazards before traversing these terrains is of paramount importance [2]. The surfaces of celestial bodies such as Mars

and the Moon exhibit deformable and soft characteristics, making them susceptible to longitudinal slip and sinkage [3]. These undesirable conditions not only compromise trajectory tracking performance and add time delays but also risk rover immobilization. This was exemplified by the incident involving the Opportunity rover in 2005, which became deeply entrenched in loose Martian soil for weeks before managing to escape itself [4], [5]. Similarly, the Spirit rover experienced permanent immobilization in 2009 [4]. The underlying cause of these accidents can be attributed to insufficient driving force of the rover wheels to overcome loose

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soil resistance, causing excessive slippage [6], [7]. Numerous researchers have dedicated efforts to developing methods for compensating for slip in planetary rovers, highlighting the complexity of managing such dynamics effectively [8], [9]. Despite these efforts, controlling slip continues to pose a considerable challenge [10]. Some researchers are employing traction control techniques to deal with slip to enhance rover navigation and safety on extraterrestrial terrains [11], [12]. The development of strategies to estimate slip and hence assess terrain traversability for the navigation algorithms [13], [14] has made pivotal contributions to rover navigation. This approach enables the maintenance of optimal traction, mitigating the risk of immobilization, and improving the mobility and safety of rovers in space exploration missions [15].

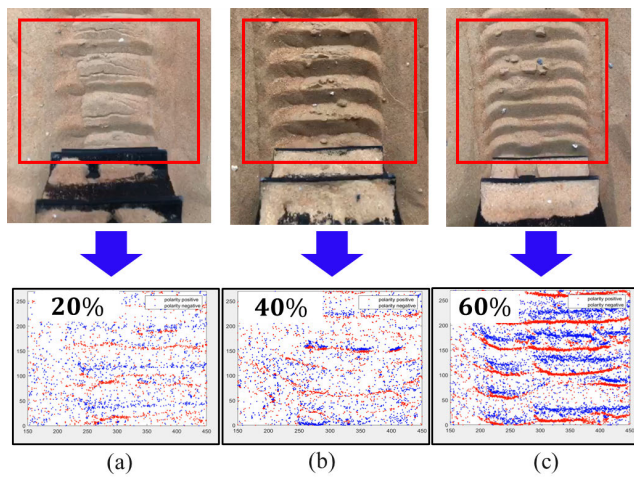


FIGURE 1. Events from a neuromorphic vision sensor capturing variations in slip conditions. (a) low slip of 20% (b) moderate slip of 40% (c) high slip of 60%.

Longitudinal slip estimation for planetary rovers is an area of active research, with methodologies generally falling into one of three categories: Model-based, Data-based, and Vision-based approaches as outlined in [16]. The Model-based approach relies on terramechanics mathematical models that simulate the interactions between the rover wheels and the terrain [17]. This approach can predict slip under known conditions but may struggle with rapid variations and uncertainties, making it less effective in unpredictable terrains [18]. The Data-based approach utilizes historical data from previous and current rover missions [16], [19]. It is effective when historical data is available; however, its applicability is limited in environments significantly different from previously encountered terrains. The Vision-based approach, the most popular method [2], employs onboard cameras to capture visual terrain information and advanced data processing techniques to estimate slip. This can be subdivided into two further categories, employing frame-based and event-based cameras. The frame-based approach can be used in a wide range of terrains, providing real-time feedback

without relying on prior terrain knowledge. Nonetheless, its efficacy is compromised in low-light conditions due to low temporal resolution and high sensitivity to motion blur, which are typical of many extraterrestrial environments [20]. Event-based techniques have emerged as a promising solution to the limitations encountered by frame-based approaches. These techniques can capture and process visual data in a manner that is less affected by lighting conditions, thereby enhancing the accuracy of slip estimation in a variety of environments. Fig. 1 illustrates an example of event data captured under different slip conditions.

In this paper, we propose event-based slip estimation techniques for planetary rovers subjected to terrain-induced longitudinal slip. Specifically, we propose an event-based framework for accurate wheel slip estimation in different lighting conditions, including low light. We present a comprehensive set of experiments to validate the proposed slip estimation framework on a single wheel test rig traveling on loose soils. These experiments were conducted under well-lit and low-light conditions. The experimental results demonstrate the capability of the proposed event-based framework to provide precise slip estimations, even in extreme low-light conditions.

A. RELATED WORK

Visual Odometry (VO) has been recognized as a primary vision-based technique for estimating rover slip, and its effectiveness has been confirmed in various studies [15], [21]. However, as noted in [22], VO comes with computational challenges, especially when the rover is in motion [23], which can significantly hinder the rover's mobility [24], [25]. Dedicated FPGA processors have been suggested to reinforce VO performance [26], but terrains with minimal visual features and variable lighting conditions present persistent challenges [27]. Another challenge is the need for rovers to periodically stop to acquire images, as indicated by [28].

Innovations in sensor technology offer promising alternatives to traditional frame-based VO methods. The introduction of Time-of-Flight (ToF) cameras is a notable advancement, enabling real-time operation [29]. While the ToF camera is less affected by lighting conditions, it cannot discern color information and may introduce errors due to camera misalignment [24]. Addressing these challenges, some researchers have proposed innovations such as vertically facedown cameras combined with LED lights [30], which could potentially mitigate the problems arising from poor lighting conditions. [31] highlighted that vision, despite its importance, can sometimes be misled by deceptive terrains. With the varied and often unpredictable space terrains [32], the challenges for rovers are further accentuated. While some rovers operate with the assumption of flat terrain, this can be problematic given Mars's landscape diversity [33]. Proprioceptive sensing integrated with vision-based techniques present a new frontier in rover localization, as explored in [34]. With Neuromorphic Vision Sensors

(NVS), a paradigm shift is possible. Unlike traditional cameras, NVS detects scene changes, providing data only when pixel intensity changes. This promises a more efficient way to estimate real-time wheel slip, even in extreme low-light conditions.

B. CONTRIBUTIONS

In this work, we advance vision-based slip estimation techniques using the neuromorphic vision sensors and present a high-fidelity and robust longitudinal slip estimation framework. The original contributions of this work are as follows:

- 1) We propose a novel event-based longitudinal slip estimation framework for improved performance in low-light conditions. This method synergistically combines a filter and a transformer for feature extraction and accurate slip estimation.
- 2) We introduce a method to generate a dataset from the real-time wheel traces for longitudinal slip estimation.
- 3) We validate the framework efficacy through single-wheel test experiments under varied lighting conditions and motions.

The article is structured as follows. **Section II** delves into the concept of an intelligent wheel system and provides an overview of the experimental setup for the single wheel test rig. **Section III** presents the novel slip estimation framework design, outlining the methods used. **Section IV** delves into the discussion of the experimental design, case study, and evaluation metrics. **Section V** details the performance evaluation of the proposed framework for estimating slip in well-lit and low-light conditions. Finally, **Section VI** provides a summary of findings and outlines potential directions for future research.

II. INTELLIGENT WHEEL SYSTEM

Fig. 2 illustrates the developed wheel system, which include: (a) the grouser wheel, (b) neuromorphic vision Sensor (NVS), and (c) force/torque (F/T) sensor. Central to the experiments is the Grouser wheel, which interacts with the terrain surface to produce distinct trace patterns called wheel traces. These traces vary under different slip conditions and provide visual cues of the wheel-terrain interaction dynamics.

The NVS, positioned behind the wheel contrary to the camera position in [29], captures the unique wheel traces. Leveraging an event-driven approach, the NVS tracks evolving patterns with high temporal precision, translating these patterns into a continuous stream of events, as shown in Fig. 1. These events combined with advanced signal processing techniques, yield real-time insights into the wheel's slip.

A. NEUROMORPHIC VISION SENSOR

A Neuromorphic Vision Sensor (NVS), also known as an Event-based Camera, detects changes in pixel intensity asynchronously and transmits data only when significant scene changes occur. This contrasts with traditional cameras

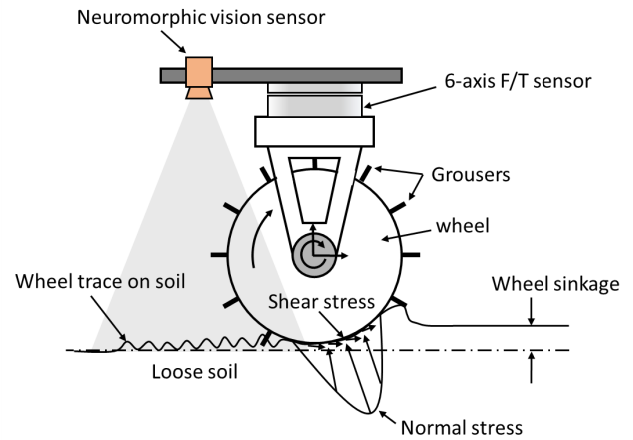


FIGURE 2. Illustration of intelligent wheel system.

that capture frames at fixed intervals. This unique system has outstanding features that include: (a) low latency of 20 μ s, (b) a temporal resolution of 800 kHz, (c) a high dynamic range (HDR) of 120 dB, (d) power efficiency of 30 mW, (e) operation in low-light conditions, and (f) an absence of motion blur [35].

1) EVENT GENERATION MODEL

The NVS operates based on a distinctive principle wherein each pixel functions independently, reacting to variations in its logarithmic photocurrent represented as $L \approx \log(I)$. Specifically, in noise-free conditions, an event denoted as $e_i = (x_i, y_i, t_i, z_i)$, where $(x_i, y_i)^T$ is the pixel location and time t_i once the change in brightness from the last event at that pixel reaches a certain threshold. This can be represented as in (1), adopted from [36]:

$$\begin{aligned} \Delta L(x_i, y_i, t_i) &= L(x_i, y_i, t_i) - L(x_i, y_i, t_i - \Delta t_i), \\ \Delta L(x_i, y_i, t_i) &= z_i \times C. \end{aligned} \quad (1)$$

where C is the contrast threshold and $C > 0$, Δt_i denotes the interval since the last event at the same pixel, and the polarity $z_i \in \{+1, -1\}$ signifies the direction of the brightness change.

2) EVENTS AND THE TEMPORAL DERIVATIVE OF BRIGHTNESS

These highlight that pixels in an event camera establish a threshold on brightness changes since the last event occurrence. For small intervals Δt_i , the brightness change can be approximated via Taylor's expansion as in (2):

$$\Delta L(x_i, y_i, t_i) \approx \frac{\partial L}{\partial t}(x_i, y_i, t_i) \times \Delta t_i. \quad (2)$$

therefore, events effectively offer a glimpse into the temporal derivative of brightness in the form of (3):

$$\frac{\partial L}{\partial t}(x_i, y_i, t_i) \approx \frac{z_i \times C}{\Delta t_i}. \quad (3)$$

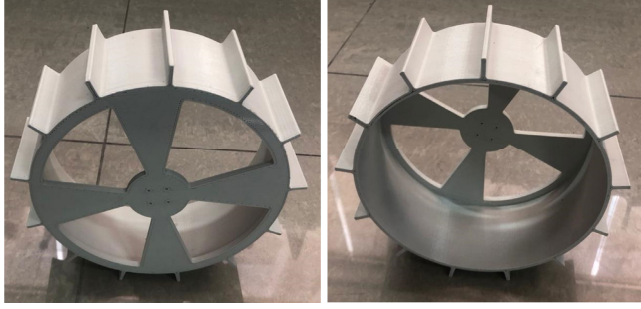


FIGURE 3. Cylindrical wheel configuration with fourteen fixed grousers.

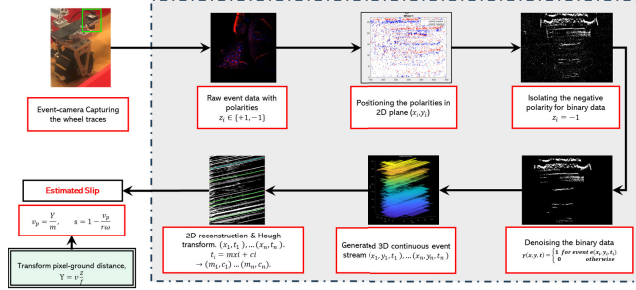


FIGURE 4. Proposed framework for the event-based slip estimation.

B. GROUSER WHEEL

This study focuses primarily on cylindrical-shaped wheels, designed to resemble the wheels of the UAE Rashid rover [37], [38]. Each of these wheels is specifically designed with 14 fixed Grousers or Lugs equally spaced as seen in Fig. 3, strategically positioned to enhance traction. The primary role of these Grousers is to improve grip, especially when navigating extremely loose soil terrains, thereby enhancing the overall traction control. The wheel specifications used in this research are as follows:

- Diameter: 0.2 m
- Width: 0.075 m
- Grousers height: 0.015 m
- Grousers width: 0.0003 m
- Weight: 1 kg

III. PROPOSED EVENT-BASED SLIP ESTIMATION FRAMEWORK DESIGN

The workflow for the proposed event-based framework for slip estimation is presented in Fig. 4. The framework begins with the NVS sensor capturing the wheel traces, then acquiring the wheel trace data via event processing, which captures the raw event data characterized by the polarity and spatial coordinates. This data is then projected onto a 2D plane for further processing. The subsequent step involves isolating the negative polarity events and converting them into a binary format, which streamlines the data for noise reduction.

A. NOISE FILTRATION

We adopt the Yang filter for designing our denoising filter [39]. Now the event data can be represented by the three

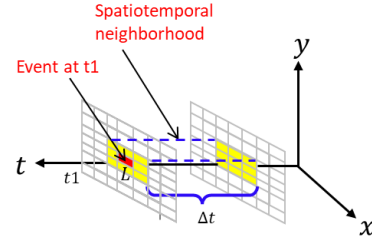


FIGURE 5. Illustration of the spatiotemporal neighborhood.

parameters $e(x, y, t)$, consisting of the spatial coordinates x, y on the image plane and a timestamp t , as demonstrated in Fig. 5. The red point marks the newly arrived event at t_1 , situated centrally within a spatial neighborhood of size $L \times L$, where L is an odd integer.

Events within this neighborhood aggregate to form a density matrix D . The element $D_{j,k}$ is given by (4):

$$D_{j,k} = \sum_{t=t_i-\Delta t}^{t_i} \gamma \left(x_i - \frac{L-1}{2} - 1 + j, y_i - \frac{L-1}{2} - 1 + k, t \right) \quad (4)$$

where x_i, y_i are the spatial coordinates of the newly arrived event, and $\gamma(x, y, t)$ is a binary function defined as:

$$\gamma(x, y, t) = \begin{cases} 1 & \text{if there is an event } e(x, y, t) \\ 0 & \text{otherwise} \end{cases}$$

NVS events can be depicted as pulse functions:

$$e_i(x_i, y_i, t_i) \Rightarrow \delta(x - x_i, y - y_i, t - t_i) \quad (5)$$

The summed event pulses are expressed as:

$$ES(x, y, t) = \sum_{i=1}^N \delta(x - x_i, y - y_i, t - t_i) \quad (6)$$

Event density, d , is represented by the L1 norm of D as:

$$d = \|D\|_1 \quad (7)$$

Real events surpass a threshold, represented as:

$$\iiint_{x_i y_i t_i \in \Omega_{\Delta t}^L} \delta(x - x_i, y - y_i, t - t_i) dx dy dt \begin{cases} < \Psi & \text{for BAs} \\ \geq \Psi & \text{for objects} \end{cases} \quad (8)$$

where the threshold $\Psi \leq 45$ and BAs is the background activities.

In the noise-reduction phase, the flicker noise decision value R is defined by:

$$R = \|\langle P, D \rangle\|_1 \quad (9)$$

where P is:

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Real events are retained if $R \neq 0$; otherwise, they are classified as noise.

B. HOUGH TRANSFORM

The filtered event data is reconstructed in 3D space (x, y, t) to extract geometrical features, specifically straight lines, for our studies. We apply the Hough Transform to map these features into a simplified parameter space known as the Hough space. The line representation in Hough space is defined by two parameters: the slope m and the t-intercept c . A straight line in the image space, represented in the cartesian coordinates (x, t) , can be defined as in (10):

$$t_i = mx_i + c_i \quad (10)$$

Through a transformation, each point in the (x, t) space can be represented as a line in the (m, c) parameter space as follows:

$$c_i = -mx_i + t_i \quad (11)$$

Given multiple points in the image space, solving for c and m will yield lines in the Hough space. The intersection of these lines in (m, c) space indicates that the corresponding points in (x, t) space lie on the same straight line. The coordinates of the intersection point provide the slope and t-intercept of this line. Utilizing this approach, we process the accumulated event data to identify the linear patterns in the (x, t) plane into (m, c) space.

C. SLIP CALCULATION

A pinhole camera model is employed to establish the geometric relationship between the terrain plane and the camera plane, enabling the computation of the distance per pixel from the event camera. This calculation is based on the projection of 3D points(world coordinates) onto the 2D image plane(camera coordinates) [40]. In our test rig, the y-axis represents the direction of motion, and the terrain surface distance per pixel Y can be computed as:

$$Y = v \left(\frac{Z}{f} \right) \quad (12)$$

where v represents each pixel value, f is the focal length of the event camera, and Z is the distance between the camera and the terrain surface. Z is a sinkage-dependent parameter. A uniform value of Z corresponding to a sinkage of 0.001m is assumed for simplicity.

$$v = f \left(\frac{Y}{Z} \right) \quad (13)$$

Now, the estimated translational velocity v_p , is mathematically modeled as:

$$v_p = \frac{Y}{m} \quad (14)$$

Integrating the estimated translational velocity v_p into (15) yields the estimated slip ratio.

$$s_p = \left(1 - \frac{v_p}{r\omega} \right) \times 100\% \quad (15)$$

where s_p denote the estimated slip ratio in (%), ω represents the angular velocity of the wheel, r is the wheel radius.

TABLE 1. Extracted camera parameters and derived values.

Parameter	Value
Focal Length (f)	758.8
Distance between Camera and Terrain (Z)	0.285 m
Pixel Distance (Y)	3.7557×10^{-4} m

IV. EXPERIMENTS

Experiments were performed on the test platform to evaluate the performance of the proposed event-based slip estimation framework on a single wheel operated on soft terrain (subjected to longitudinal slip) under different lighting conditions.

A. TEST PLATFORM

A single-wheel test rig, illustrated in Fig. 6, is employed for the experimental phase of this work. This custom-designed test rig facilitates comprehensive investigations into a wheel traversing soft terrain(loose soil). It has various components, including an intelligent wheel system, a soil bin with dimensions of 1.5 m in length, 0.9 m in width, and 0.4 m in depth, a mechanical frame, safety equipment, sensors, actuators, and a control box. The test rig incorporates two motors: a DC motor with a planetary drive gear for wheel operation in closed-loop mode for rotational speed control and a stepper motor for carriage control to regulate slip conditions. Table 2, provides the essential features of these various components. Moreover, in our test rig, the wheel motor is controlled using a DC motor controller enclosed in the main control unit, maintaining a constant wheel rotational velocity based on the Proportional Integral and Derivative (PID) control algorithm, with tuned gains of ($K_p = 1.2$, $K_i = 1.7$, $K_d = 0.001$). The carriage motor ensures a constant horizontal traveling velocity. This test rig is designed to accommodate a wheel weight of 60 N and to travel(forward/backward) a distance of 1.4 m across the loose soil.

The soil sample employed in the experiments is highly loose and deformable, representing the attributes of the south-pole lunar terrain. Additionally, the soil properties were obtained to estimate the anticipated slip, and to develop a virtual simulation model for the wheel-terrain interactions [38]. Table 3, provides a summary of the soil parameters.

Moreover, the wheel motor is equipped with an absolute encoder to precisely measure the rotational velocity, while the carriage motor operates at a fixed translational velocity. By combining these measurements, the actual slip for the

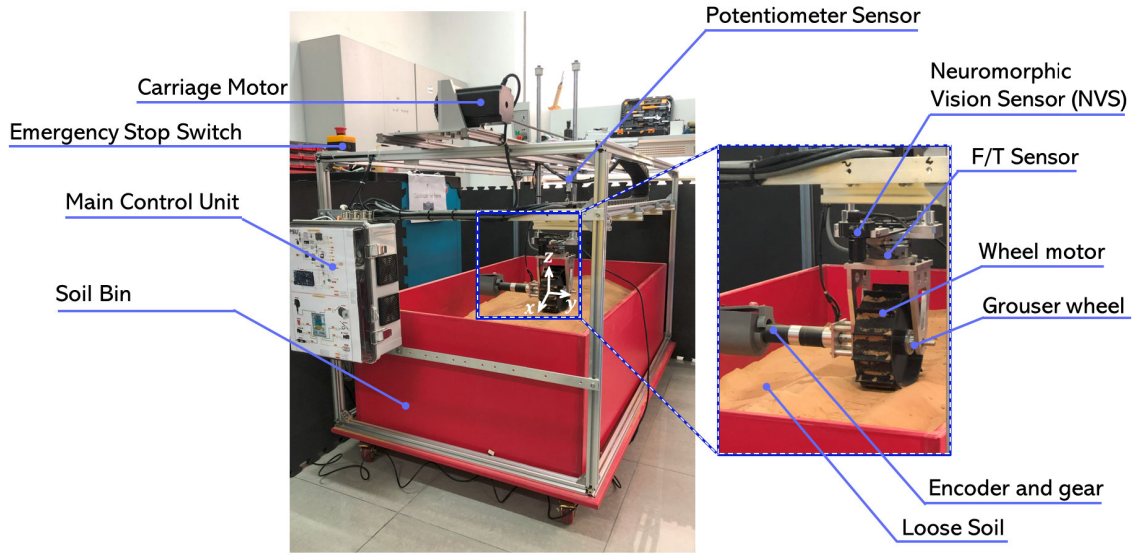


FIGURE 6. A detailed view of the single-wheel test rig, showcasing the wheel traversing loose soil.

TABLE 2. Essential features of test rig components.

Components	Parameters	Value	Units
Wheel Motor (MMP S17-400D-24V GP42-408 DC gearmotor)	Input voltage	24	VDC
	Typical rated current	3.1	A
	Rated speed	4335	RPM
	Rated torque	16	oz-in
	Gearhead-ratio	407-1	
Carriage Motor (DM8601 Stepper Motor)	Output peak current	2.4 - 7.2	A
	Typical Input voltage	68	VDC
	Logic Signal Current	7 - 16	mA
	Pulse input frequency	0 - 200	kHz
	Min Pulse Width	2.5	μ S
	Number of micro-step resolutions	16	
Wheel Motor Encoder (E40H8-1024-6-L-5 Absolute encoder)	Max. response frequency	300	kHz
	Max. allowable revolution	5000	RPM
	Resolution	1024P/R	

ground truth of the wheel trajectory can be mathematically computed as in [38].

$$s_a = \left(1 - \frac{v_a}{r\omega}\right) \times 100\% \quad (16)$$

where s_a denotes the actual slip ratio in (%), and v_a denotes the actual translational velocity of the carriage motor. Besides, we only care about the case of $r\omega \geq v_a$ in (16), corresponding to $s_a \geq 0$. This aims to replicate some of the real longitudinal slips encountered by planetary rovers in space environments [8], [10].

B. COMPARISON BETWEEN TEST PLATFORM AND ACTUAL ROVER CONDITIONS

To ensure our test platform closely emulates the actual operating environment of a rover, several key design considerations were implemented:

TABLE 3. Collected soil parameters for the lunar terrain.

Parameter	Value	Unit
k_c'	0.63	-
k_ϕ'	80	-
K	0.036	m
n	1.18	-
γ	15620	N/m ³
$d\gamma$	$0.1 \times \gamma$	N/m ³
η	1.15	-
ϕ	23	deg
c	0	kPa
h_b	0.01	m
C_f	800	Ns/m
k_g'	0.07	-
k_0'	0.035	-

- 1) In terms of environmental conditions: (a) the soil bin used in our experiments contains soft terrain, a highly loose and deformable soil confirmed by the Mohammad Bin Rashid Space Center (MBRSC) to be similar to the south-pole lunar terrain [37], and (b) the lighting conditions are similar to the actual space illumination range, starting from 0.5 Lux to 30 Lux.
- 2) In terms of operational conditions: (a) the wheel structure used in our tests is a replica of the UAE Rashid rover's wheel, including the 14 fixed grousers, wheel motor, wheel size, and weight, and (b) the wheel speeds are similar to the actual operational speeds of the UAE Rashid Rover, ranging from 0.02 m/s to 0.1 m/s [38].

Thus, this test platform emulates the motion dynamics and slip conditions encountered by planetary rovers.

C. EXPERIMENTAL STUDY

In this section, a series of experimental tests were conducted to validate the proposed event-based slip estimation framework under different lighting conditions, measured

in lux ranging from 0.5 to 30 lux. The slip ratio for each experiment was calculated using the mathematical description provided in (16). We designed our experiments to cover a comprehensive range of slip ratios to reflect potential real-world conditions encountered by rovers. The experiments were conducted under controlled conditions to ensure repeatability and accuracy. The detailed summary of the experimental configurations is outlined below:

- C1 - Wheel speed is fixed at 0.02 m/s, and the carriage motor is set to give a slip ratio of range $s = 2\% - 90\%$ under 30 lux
- C2 - Wheel speed is fixed at 0.04 m/s, and the carriage motor is set to give a slip ratio of range $s = 2\% - 90\%$ 30 lux.
- C3 - Wheel speed is fixed at 0.06 m/s, and the carriage motor is set to give a slip ratio of range $s = 2\% - 90\%$ 30 lux.
- C4 - Wheel speed is fixed at 0.08 m/s, and the carriage motor is set to give a slip ratio of range $s = 2\% - 90\%$ 30 lux.
- C5 - Wheel speed is fixed at 0.1 m/s, and the carriage motor is set to give a slip ratio of range $s = 2\% - 90\%$ 30 lux.
- C6 - Wheel speed is fixed at 0.04 m/s, and the carriage motor is set to give a slip ratio of range $s = 2\% - 90\%$ 0.5 lux.

These experiments were carefully designed to test the proposed framework's robustness over a range of speeds and lighting conditions that a rover might encounter on lunar surfaces. The lux levels were selected to emulate the lighting transitions from day to night and the shadowed regions that rovers frequently navigate. This extensive testing protocol ensures that the proposed framework can accurately estimate slip ratios in conditions that closely simulate those on extraterrestrial terrains [7].

D. EVALUATION METRICS

Each test was repeated three times to check the robustness of the framework in estimating slip. To quantify the effectiveness of our proposed framework, we will utilize two performance evaluation metrics: Standard Deviation (SD) and Mean Absolute Error (MAE). These metrics are defined mathematically as follows [3]:

$$SD = \sqrt{\frac{1}{n} \sum_{i=1}^n (s_{p,t} - \mu)^2} \quad (17)$$

$$MAE = \frac{1}{n} \sum_{t=1}^n |s_{a,t} - s_{p,t}| \quad (18)$$

where n is the number of observations, $s_{a,t}$ is the actual slip at instant t , and $s_{p,t}$ is the estimated slip at same instant t , and μ is the mean value.

V. RESULT AND DISCUSSION

A series of experimental tests, as outlined in the previous section, were conducted to collect data and evaluate the

performance of the proposed event-based slip estimation framework. Figure 7 illustrates the event outputs for the varied slip condition experiments under varying lighting conditions. A detailed summary of the experimental outcomes are outlined below:

A. WELL-LIT CONDITION

Five experimental tests C1, C2, C3, C4, and C5 were conducted during daylight hours with an approximate light intensity of 30 Lux. These tests had varied wheel speeds of 0.02 m/s, 0.04 m/s, 0.06 m/s, 0.08 m/s, and 0.10 m/s, encompassing different scenarios across a slip range from 2% to 90%.

Figure 8 illustrates the performance evaluation of the estimated slip compared to the actual slip for C1; an additional error bar graph is also presented. It is evident that the estimated values closely align with the actual values, with a slight deviation observed at higher slip conditions. This deviation can be attributed to the variability in sinkage during different slip conditions, affecting the parameter Z used in equation (13). The Z used for all scenarios is for 50% slip, therefore the results are higher than the actual for lower slip and it is lower than the actual for higher slip ratios. Moreover, to quantitatively assess this deviation, the Mean Average Error (MAE) and Standard Deviation (SD), are computed and presented in Table 4. It is evident from these error values that the highest MAE obtained was 4.36% for the slip ratio of 90%, indicating a relatively accurate slip estimation. In addition, the SD varied between 0.21% and 1.66% which indicates that the experiments are giving similar results for the same condition and this test is robust.

Similarly, Fig. 9 - 12 presents the performance comparison of the estimated slip and actual slip for C2, C3, C4, and C5, respectively. While a close correlation was observed in all the cases as well, the deviations were slightly higher compared to the C1. Table 4. presents the MAE errors for all the configurations. In all cases, the highest errors MAE computed were 4.95% and 4.97%, respectively, still indicating a generalization in accurately estimating slip ratio. The SD is extremely low with a maximum value recorded of 2.02% across all the tests. However, in configurations C4 and C5, the error becomes relatively higher, indicating that higher speeds are prone to low accuracy in estimating slip. For instance, the MAE of 4.95% is observed at speed 0.1 m/s and slip 70%, and a corresponding SD of 4.36% is recorded, which is still low and indicates the robustness of the developed framework.

While C4 maintains consistent estimation across the tested range in Table 4, despite a general trend of higher absolute errors. Notably, configurations C2 to C5 demonstrate enhanced accuracy at higher slip ratios (80% to 90%), suggesting robustness in scenarios of substantial slippage. However, there is a general trend across all configurations to underestimate the slip ratio as the actual slip ratio increases, except for C1 and C2 at a 2% slip ratio where the estimation is slightly higher. The most precise estimation occurs in C3

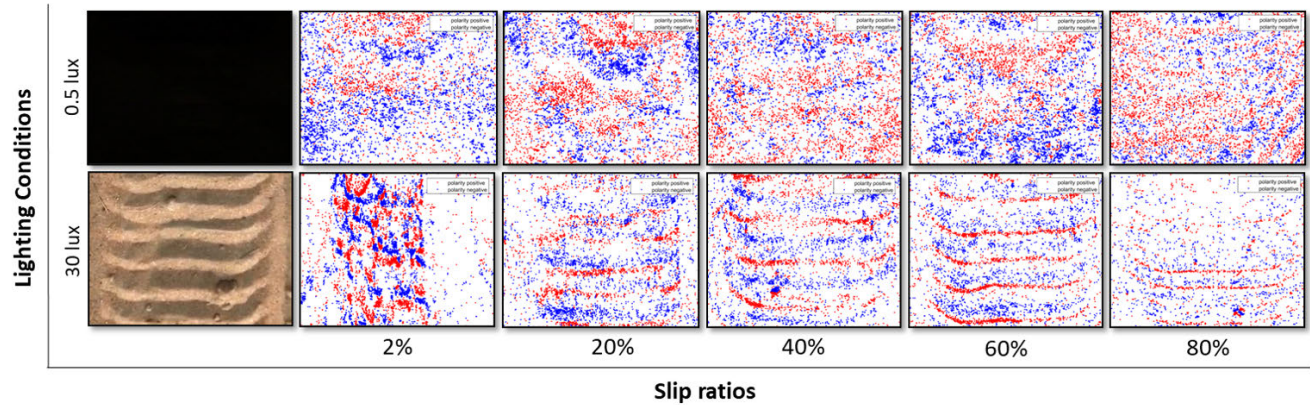


FIGURE 7. Experimental tests conducted under different lighting conditions and slip ratios.

TABLE 4. Error metrics for C1, C2, C3, C4, and C5 under 30 lux light intensity.

Actual slip, s_a (%)	Estimated Slip, s_p (%)					Standard Deviation, SD (%)					Mean Absolute Error, MAE (%)				
	C1	C2	C3	C4	C5	C1	C2	C3	C4	C5	C1	C2	C3	C4	C5
2.00	0.90	-0.85	3.03	5.21	5.21	1.18	1.35	1.58	2.53	1.75	1.10	2.85	1.03	3.21	0.67
10.00	12.54	11.99	11.81	11.75	11.75	0.60	0.00	0.00	0.00	0.68	2.54	1.99	1.81	1.75	0.24
20.00	21.24	24.66	21.02	20.10	20.10	1.56	0.00	2.02	0.02	0.52	1.24	4.66	1.02	0.10	3.71
30.00	30.30	31.56	29.51	33.86	33.86	1.14	0.02	0.00	0.00	0.00	0.30	1.56	0.49	3.86	0.60
40.00	37.40	37.25	39.97	38.42	38.42	1.22	0.02	1.18	1.99	4.36	2.60	2.75	0.03	1.58	1.88
50.00	48.63	47.83	48.56	47.14	47.14	1.66	0.00	1.11	0.00	1.32	1.37	2.17	1.44	2.86	3.25
60.00	56.29	56.38	55.78	56.00	56.00	0.21	0.00	0.60	0.00	0.00	3.71	3.62	4.22	4.00	2.29
70.00	65.73	65.75	65.03	65.34	65.34	0.45	0.00	0.00	0.48	0.07	4.27	4.25	4.97	4.66	4.95
80.00	76.18	75.43	75.70	75.74	75.74	1.21	0.01	0.40	0.33	1.26	3.82	4.57	4.30	4.26	4.10
90.00	85.64	86.57	87.16	87.25	87.25	1.15	0.01	0.34	0.00	0.10	4.36	3.43	2.84	2.75	2.05

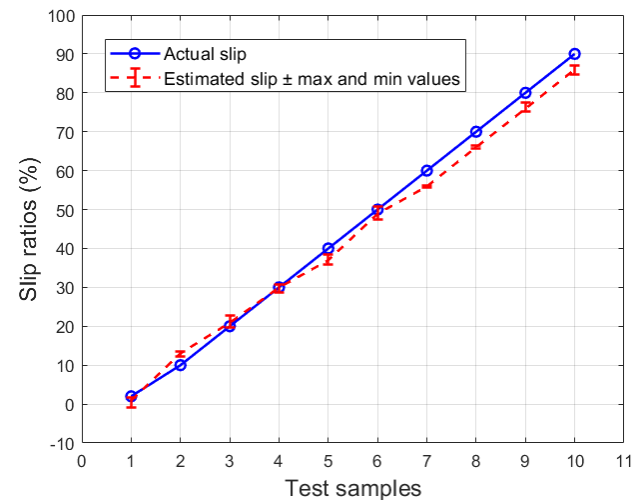


FIGURE 8. C1 - Performance comparison for wheel speed of 0.02 m/s with varied slip conditions under 30 lux light intensity.

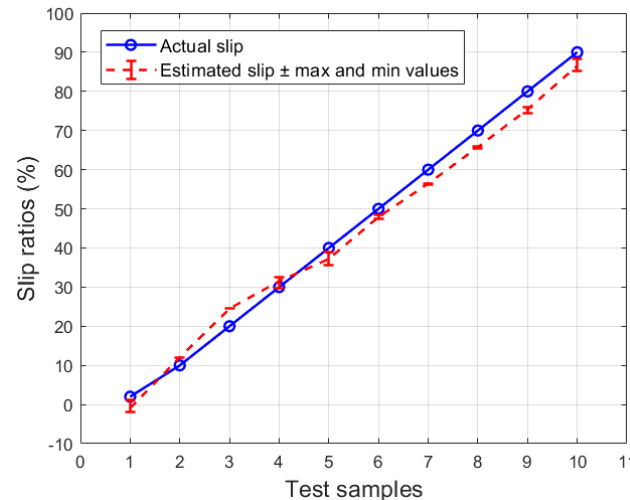


FIGURE 9. C2 - Performance comparison for wheel speed of 0.04 m/s with varied slip conditions under 30 lux light intensity.

at a 40% actual slip ratio with an MAE of 0.03%, while the least accurate is observed in C5 at 70%, with an MAE of 4.95%, indicating potential reliability issues at higher speeds and higher slip conditions for this configuration. The

overall trend across all configurations indicates increasing MAE with increasing slip ratio, which is a common issue in slip estimation due to the non-linearities and complexities involved in higher slip scenarios. The relatively lower errors

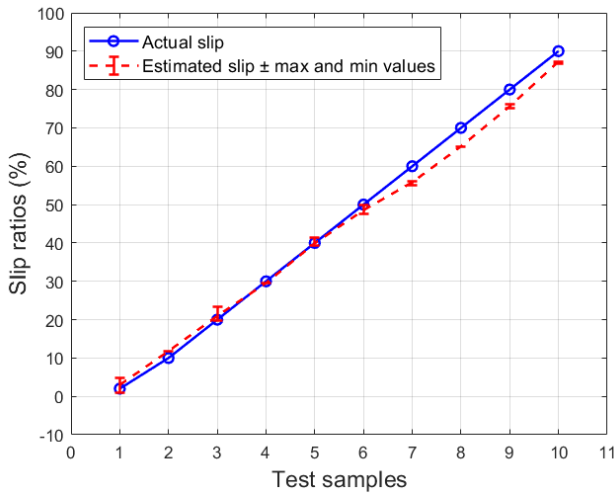


FIGURE 10. C3 - Performance comparison for wheel speed of 0.06 m/s with varied slip conditions under 30 lux light intensity.

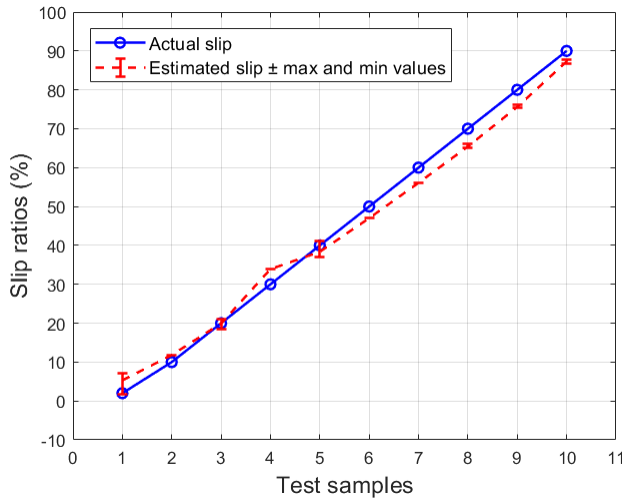


FIGURE 11. C4 - Performance comparison for wheel speed of 0.08 m/s with varied slip conditions under 30 lux light intensity.

at medium slip ratios suggest that the framework is capturing the slip dynamics effectively, due to the single-wheel test rig vibration at high speeds. Thus, we can conclude that our proposed framework consistently demonstrates accurate slip estimation across various wheel speeds in well-lit conditions.

B. LOW-LIGHT CONDITION

Another experimental test C6 was conducted during the dark hours with an approximate light intensity of 0.5 Lux, similar to Moonlight. In these experiments, we used a wheel speed of 0.04 m/s, which exhibited the moderate speed condition, encompassing different scenarios across a slip range from 2% to 90%.

Figure.13 depicts the performance comparison of the estimated slip compared to the actual slip, in both well-lit and low-light conditions. It is evident from the graph that both estimated values closely align with the actual values.

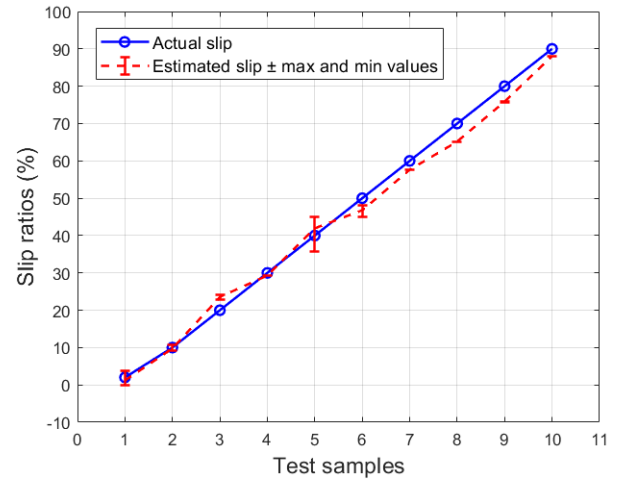


FIGURE 12. C5 - Performance comparison for wheel speed of 0.10 m/s with varied slip conditions under 30 lux light intensity.

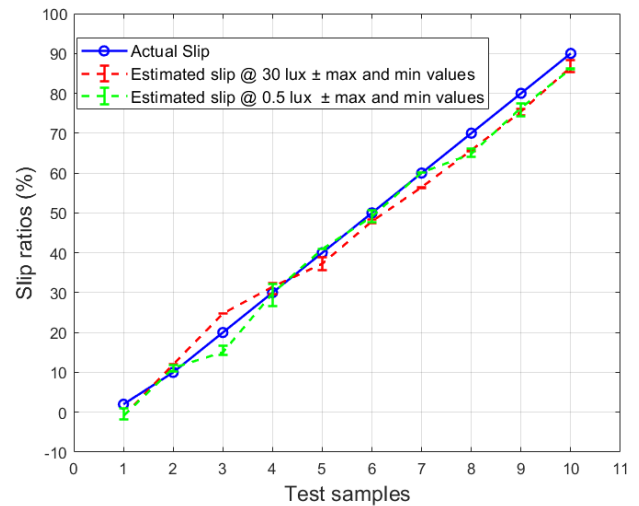


FIGURE 13. C6 - Performance comparison for wheel speed of 0.04 m/s with varied slip conditions under both 30 and 0.5 lux light intensity.

However, it is clear that the estimated values in low-lighting conditions deviate higher than the estimated values in well-lit conditions, this demonstrates the difficulty in estimating slip in extreme low-light conditions. Surprisingly, MAE in both cases is approximately the same in Table.4. This shows that the cumulative performance of the framework in both lighting conditions is almost the same. It is observed that from the error values in Table 5. the highest MAE error recorded was 5.31% in the slip of 70%, further indicating a less precision of slip estimation than the corresponding MAE of 4.66% observed in well-lit conditions for the same wheel speed and slip. Furthermore, the worst SD was observed at the lowest slip ratio, of around 1.35%. This indicates the estimation accuracy of the framework at extremely low slip fall short.

C. COMPARISON WITH OTHER APPROACH

Subsequently, we benchmarked our event-based slip estimation framework against the State-of-the-art approach using

TABLE 5. Error metrics for C6 under 0.5 lux light intensity.

Actual slip, s_a (%)	Estimated slip, s_p (%)	Standard Deviations, SD (%)	Mean Absolute Error, MAE (%)
2	-0.85	1.35	2.85
10	11.20	0.65	1.20
20	15.15	1.07	4.85
30	29.88	2.36	0.12
40	40.93	0.00	0.93
50	49.04	1.26	0.96
60	60.10	0.00	0.10
70	64.69	1.02	5.31
80	76.27	1.44	3.73
90	86.08	0.18	3.92

TABLE 6. Absolute error metrics with state-of-the-art approaches.

Actual Slip, s_a (%)	Proposed Framework MAE (%)	Higa et al. [29] MAE (%)
Low Slip, 0%	1.10	1.40
Moderate Slip, 40%	2.60	3.40
High Slip, 80%	3.82	3.90

the MAE metric. Table 6. provides the performance of our framework with that of approaches in [29] at the same speed of 0.02 m/s and light conditions across different slip ratios. Our framework consistently outperforms the competition, achieving less estimation error compared with the other approach. For instance, at a slip ratio of 0%, the error difference is 0.3%. Similarly, for 40% and 80%, the differences are 0.8% and 0.02%, respectively. However, it can be observed that at a higher slip ratio, our framework and their approach yield almost the same performance.

VI. CONCLUSION

In conclusion, this paper has presented a comprehensive study on the challenges of terrain-induced slippage for space rovers and proposed a novel event-based framework to estimate slips with high accuracy under various lighting conditions. Our experimental results, conducted under both well-lit and low-light conditions, demonstrate the framework's high accuracy of estimation across various slip conditions, with a maximum MAE of 4.66% and 5.31%, respectively. This confirmed its efficacy with an estimation accuracy consistently above 94%. Moreover, the demonstrated maximum SD of 1.35% underlines the framework's reliability. These results represent a noteworthy advancement in navigation systems for rovers traversing soft terrains, particularly for low-speed operations. **When contrasted with current state-of-the-art methods, our framework shows superior performance in low to moderate slip conditions.** Nevertheless, there remains scope for further improvement at high slip conditions. In addition, this is a standard preliminary test for all rovers, as it stimulates only the wheel of the rover, not the whole rover.

Furthermore, in our test study, a 14-grouser wheel was used; however, the approach is applicable to wheels with any number of grousers. Additionally, with the camera mounted on the wheel's rear, there is a delay in capturing the wheel's past path, which may introduce a lag in slip estimation. Future work should aim to refine our framework to enable real-time slip estimation and adapt to wheel designs with no grousers.

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